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Polarity control in a single transition metal dichalcogenide (TMD) transistor for homogeneous complementary logic circuits†

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Recently, there have been various attempts to demonstrate the feasibility of transition metal dichalcogenide (TMD) transistors for digital logic circuits. A complementary inverter circuit, which is a basic building block of a logic circuit, was implemented in earlier works by heterogeneously integrating n- and p-channel transistors fabricated on different TMD materials. Subsequently, to simplify the circuit design and fabrication process, complementary inverters were constructed on single-TMD materials using ambipolar transistors. However, continuous transition from the electron-conduction to the hole-conduction state in the ambipolar devices led to the problem of a high leakage current. Here, we report a polarity-controllable TMD transistor that can operate as both an n- and a p-channel transistor with a low leakage current of a few picoamperes. The device polarity can be switched simply by converting the sign of the drain voltage. This is because a metal-like tungsten ditelluride (WTe₂) with a low carrier concentration is used as a drain contact, which subsequently allows selective carrier injection at the palladium/tungsten diselenide (WSe₂) junction. In addition, by using the operating principle of the polarity-controllable transistor, we demonstrate a complementary inverter circuit on a single TMD channel material (WSe₂), which exhibits a very low static power consumption of a few hundred picowatts. Finally, we confirm the expandability of this polarity-controllable transistor toward more complex logic circuits by presenting the proper operation of a three-stage ring oscillator.

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Introduction

Since the experimental advent of graphene in 2004, this two dimensional material has received considerable attention owing to its superior electrical and mechanical properties.^{1,2} However, the absence of an energy bandgap has hindered the

development of graphene-channel transistors toward logic circuits, because a low energy bandgap of channel materials typically induces a huge leakage current.³ In this regard, another type of atomically thin layered material, transition metal dichalcogenides (TMDs), has been proposed for the realization of future nanoscale devices.^{4,5} Most TMD materials are generally suitable for use as the channels of transistors because they have finite bandgaps between 1.0 and 2.0 eV for normal transistor operation.⁶ Under this technical circumstance, there have been various experimental attempts to demonstrate the feasibility of TMD transistors for digital logic circuits. A complementary inverter circuit, which is a basic building block of a logic circuit, was implemented in previous works by heterogeneously integrating n- and p-channel transistors on two different TMD materials.^{7–9} Subsequently, to simplify the circuit design and fabrication process, complementary inverters have been constructed on single-TMD materials using ambipolar transistors.^{10–16} Das *et al.* combined WSe₂ n- and p-channel transistors using a multiple-gate structure,¹² where the n- and p-channels in single WSe₂ material were indepen-

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dently controlled by two gate electrodes. However, this device required two more terminals compared to conventional inverter circuits, thereby increasing the complexity of the fabrication process. Lin *et al.*¹⁴ and Lim *et al.*¹³ demonstrated complementary inverters by integrating two ambipolar transistors with a common gate on single molybdenum ditelluride (MoTe_2) material, but continuous transition from the electron-conduction to the hole-conduction state in the ambipolar devices led to a high leakage current.

Here, we report a polarity-controllable TMD transistor that can operate as both an n- and a p-channel transistor with a low leakage current of a few picoamperes ($<10 \text{ pA } \mu\text{m}^{-1}$). The polarity of the device is switched simply by converting the sign of the drain voltage (V_{DS}). This is because the metal-like tungsten ditelluride (WTe_2), with a low carrier concentration, is used as a drain contact, which subsequently allows selective carrier injection at the palladium (Pd)/tungsten diselenide (WSe_2) junction. To understand the carrier-transport mechanism in the polarity-controllable transistor, we perform various analyses, including Hall effect measurement, Kelvin probe force microscopy (KPFM), and temperature-dependent electrical measurement. In addition, by using the operating principle of the polarity-controllable transistor, we demonstrate a complementary inverter circuit on a single TMD channel material (WSe_2), which exhibits a very low static power consumption of a few hundred picowatts. Finally, we demonstrate a three-stage ring oscillator consisting of the inverters based on the polarity-controllable transistor.

Experimental

Fabrication of the polarity-controllable transistor

A WSe_2 flake was mechanically exfoliated onto a 90 nm-thick $\text{SiO}_2/\text{p}^+\text{-doped Si}$ substrate using adhesive tape (244SPV, Nitto). To remove the tape residue, the WSe_2 sample was rinsed with acetone for 1 h. Next, a WTe_2 flake was exfoliated onto a poly(methyl methacrylate) (PMMA)/poly(vinyl alcohol) (PVA)/Si substrate. This WTe_2 sample was then dipped in de-ionized (DI) water and the PVA layer was dissolved, subsequently floating a WTe_2/PMMA layer on the surface of the DI water. The floating WTe_2/PMMA layer was moved onto a metal slide and then transferred onto the $\text{WSe}_2/\text{SiO}_2/\text{Si}$ substrate through a mechanical transfer process. After that, the PMMA layer was removed using acetone, leaving $\text{WSe}_2/\text{WTe}_2$ stack on the SiO_2/Si substrate. Finally, to form source and drain electrodes, Pd/Au (10/20 nm) layers were deposited in an electron-beam evaporation system, followed by a lift-off process.

Fabrication of the $\text{WTe}_2/\text{WSe}_2/\text{WTe}_2$ transistor

A WSe_2 flake was exfoliated onto a 90 nm-thick SiO_2 oxide layer on heavily doped Si substrate using adhesive tape. Then, WTe_2 flakes were stacked on two edges of the WSe_2 flake using the mechanical transfer process as explained earlier. The metal contact patterns were formed on the WTe_2 flakes by a photo-

lithography process, followed by Pd/Au (10/30 nm) deposition process *via* an e-beam evaporator.

Characterization of WSe_2 and WTe_2

The KPFM measurement was performed using NTEGRA Spectra (NT-MDT). The Hall measurement was conducted using a Keithley 2182A to apply constant excitation current and a Keithley 6221 for measuring the Hall voltages. A magnetic field with magnitude from 0 T to 0.36 T was perpendicularly applied to the sample.

Electrical characterization of the polarity-controllable transistor

The electrical transport measurements were conducted with a probe station using a Keysight B2912A semiconductor parameter analyzer at room temperature. The temperature-dependent electrical characteristics were measured using a Keithley 4200 semiconductor parameter analyzer in a vacuum chamber (below 10^{-4} Torr).

Fabrication of the complementary inverter

A WSe_2 flake was first exfoliated onto a 90 nm-thick SiO_2/Si substrate. Then, a WTe_2 flake was stacked onto the WSe_2 flake using a mechanical transfer process as explained earlier. The metal electrode regions were defined through a photolithography process. Pd/Au (10/30 nm) layers were deposited *via* e-beam evaporation to form the contacts for WTe_2 and WSe_2 , followed by a lift-off process in acetone. The supply voltage (V_{DD}) was applied to one of the electrodes on WSe_2 , and the other electrode on WSe_2 was connected to the ground (V_{SS}). The common back gate served as the input voltage (V_{IN}) electrode and the output voltage (V_{OUT}) was measured at the electrode on WTe_2 .

Results and discussion

WTe_2 is known to distort octahedral coordination ($1\text{T}'$) structures and exhibit semi-metallic electronic properties.^{17,18} This metallic phase facilitates the use of WTe_2 as an electrode for the transistor. We fabricated a $\text{WSe}_2/\text{WTe}_2$ heterojunction transistor using WTe_2 as an electrode, as shown in the schematic illustration of Fig. 1a and the optical microscope image of Fig. 1b. The thickness of the WSe_2 and WTe_2 layers were confirmed to be approximately 8 nm and 30 nm by atomic force microscope (AFM), respectively (Fig. S1†). Here, the Pd electrodes on WSe_2 were labeled as M_1 and M_2 , and the electrodes on WTe_2 were labeled as M_3 and M_4 . Before analyzing the $\text{WSe}_2/\text{WTe}_2$ heterojunction transistor, we first investigated the electrical characteristics of WSe_2 and WTe_2 transistors. Fig. 1c shows the drain current (I_{DS})–gate voltage (V_{GS}) characteristics at $V_{\text{DS}} = -3 \text{ V}$ for WSe_2 (M_1 : source and M_2 : drain) and WTe_2 (M_3 : source and M_4 : drain) transistors. Ambipolar behavior was observed in the WSe_2 transistor (green dotted line in Fig. 1c). This indicates that the Fermi level of Pd was pinned closely to the middle of the WSe_2 energy bandgap and, sub-

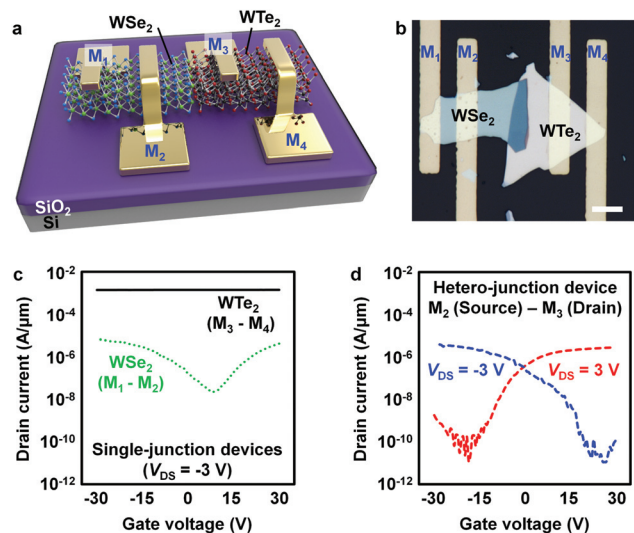


Fig. 1 (a) Three-dimensional schematic illustration of the WSe₂/WTe₂ heterojunction transistor. (b) Optical image of the WSe₂/WTe₂ heterojunction transistor. The scale bar represents 10 μm. (c) Drain current–gate voltage (I_{DS} – V_{GS}) characteristic curves of the WTe₂ (black solid line) and WSe₂ (green dotted line) transistors, where $V_{DS} = -3$ V. (d) I_{DS} – V_{GS} characteristic curves of the WSe₂/WTe₂ heterojunction transistor at $V_{DS} = 3$ V (red dashed line) and $V_{DS} = -3$ V (blue dashed line).

sequently, similar Schottky barrier heights were formed for electron and hole carriers.¹⁹ Meanwhile, the drain current in the WTe₂ transistor was hardly affected by the applied V_{GS} and also exhibited a significantly high value above 10^{-3} A μm⁻¹ (black solid line in Fig. 1c). This value is similar to or slightly higher than that obtained for graphene.³ It is because the bottom of the conduction band and the top of the valence band in the Brillouin zone overlap.²⁰ We then investigated the electrical characteristics of the WSe₂/WTe₂ heterojunction transistor (M₂: source and M₃: drain in Fig. 1d). Interestingly, the WSe₂/WTe₂ heterojunction transistor exhibited n-type transport behavior for a positive V_{DS} (red dashed line in Fig. 1d), and p-type behavior for a negative V_{DS} (blue dashed line in Fig. 1d). For reference, similar electrical characteristics were observed in three different WSe₂/WTe₂ heterojunction transistors (Fig. S2†). This means that one heterojunction transistor can work as both types of transistor by converting simply the sign of V_{DS} . In particular, unlike typical ambipolar transistors, which cannot be completely turned off owing to their continuous transition from the electron-conduction to the hole-conduction state, this polarity-controllable transistor showed low leakage currents below 1 nA/μm at -15 V < V_{GS} < -30 V for the n-type regime and at 15 V < V_{GS} < 30 V for the p-type regime.

As shown in Fig. 2a, WTe₂ presented metallic behavior but had a lower conductivity (0.40×10^2 S cm⁻¹) than Pd (2.99×10^2 S cm⁻¹) (the positive temperature coefficient of resistivity in WTe₂ can be found in Fig. S3†). Here, the electrical conductivities of WTe₂ and Pd were extracted from the slope of the I – V curves. To understand the origin of low conductivity of WTe₂, we then performed Hall effect measurements. First, the Hall voltage (V_{Hall}) was measured and then we extracted the Hall

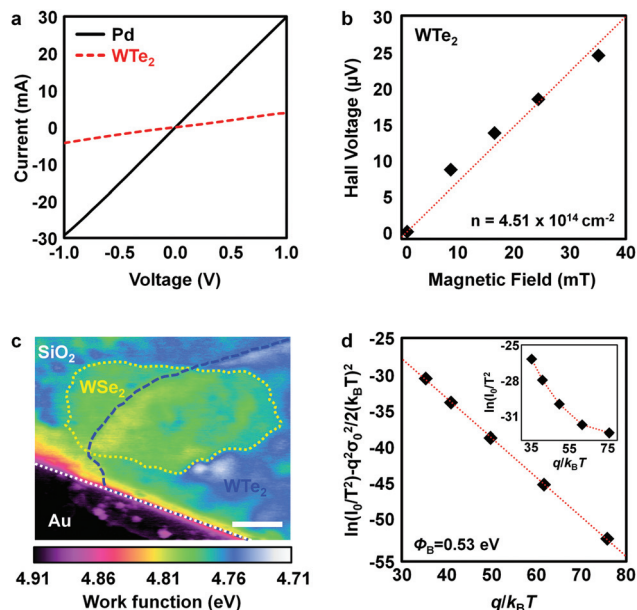


Fig. 2 (a) Electrical characteristics of Pd (black solid line) and WTe₂ (red dashed line). (b) Measured Hall voltage of WTe₂ with respect to the magnetic field. (c) Work function mapping image of the WSe₂/WTe₂ heterojunction, which was obtained by KPFM analysis. The scale bar represents 5 μm. (d) Modified Richardson plot of the Pd/WSe₂/Pd TFT. The inset shows a conventional Richardson plot of the Pd/WSe₂/Pd TFT.

coefficient ($R_H = R_{xy}/B$) at room temperature, where R_{xy} is the Hall resistance and B is the applied magnetic field. The R_H was determined by linearly fitting the slope of the R_{xy} – B graph. The carrier concentration (n) of WTe₂ obtained from $n = 1/(eR_H)$ was 1.50×10^{20} cm⁻³, where e is the electron charge. This value is significantly higher than the carrier concentrations reported in typical TMD materials (10^{15} – 10^{19} cm⁻³)²¹ and lower than the free-electron concentrations in metals (10^{22} – 10^{23} cm⁻³).²² Thus, when WTe₂ is used as an electrode in thin-film transistors (TFTs), it is expected that the carrier injection to be relatively suppressed, compared with those using Pd. KPFM measurement was also conducted to investigate the band alignment of the WSe₂/WTe₂ heterojunction. Fig. 2c shows the KPFM mapping image of the WSe₂/WTe₂ heterostructure. Before the KPFM measurement, the KPFM tip (Pt/Ir-coated Si tip) was calibrated on a highly oriented pyrolytic graphite (HOPG) surface, on which its work function is well known to be 4.6 eV.²³ Then, the work function of the KPFM tip (Φ_{tip}) was determined from the sum of the HOPG work function (Φ_{HOPG}) and the contact potential difference (ΔV_{CPD}) between the KPFM tip and the HOPG surface. By measuring the ΔV_{CPD} values between the tip and the samples (WSe₂ and WTe₂), the work function values (Φ_s) of WSe₂ and WTe₂ were calculated using the following equation: $\Phi_s = \Phi_{tip} - \Delta V_{CPD}$. The average work function values of the WSe₂ and WTe₂ films were estimated to be approximately 4.78 and 4.75 eV, respectively. In addition, we found the barrier height between Pd and WSe₂ through temperature-dependent electrical measurements in the WSe₂ transistor with Pd source/drain

electrodes. Because the WSe₂ transistor operates under thermionic emission and diffusion mechanisms, a conventional Richardson plot does not give a linear fit over the whole temperature range (inset in Fig. 2d). Thus, we employed a modified Richardson plotting method based on a thermionic emission–diffusion model.^{24,25} As shown in Fig. 2d, the linearity in the $\ln(I_0/T^2) - q^2\sigma_0^2/2(k_B T)^2$ versus $q/k_B T$ plot was obtained, where I_0 is the saturation current, T is the measurement temperature, q is the elementary charge, σ_0 is the standard deviation of the Gaussian distribution, and k_B is the Boltzmann constant. From the slope in the modified Richardson plot, the hole barrier height between WSe₂ and Pd was estimated to be 0.58 eV.

Based on the obtained work function and barrier height values, an energy band diagram before contact for Pd, WSe₂, and WTe₂ is predicted in Fig. 3a. The band properties for WSe₂, such as electron affinity and energy bandgap, were obtained from the literature.²⁶ Fig. 3b shows the predicted energy band diagram under the equilibrium state after contact-

ing Pd, WSe₂, and WTe₂ (Pd/WSe₂/WTe₂ heterojunction). When determining the Pd/WSe₂ and WSe₂/WTe₂ junctions, we ignored the Fermi level pinning effect, consequently aligning the Fermi levels of Pd and WTe₂ to the middle of the WSe₂ energy bandgap. We then analyzed the operating principle of the polarity-controllable transistor using WTe₂ as the drain electrode and Pd as the source electrode, as shown in Fig. 3c. Under negative- V_{DS} and positive- V_{GS} conditions (top-left diagram in Fig. 3c), the injections of hole and electron carriers into WSe₂ are expected to be suppressed. This is because of the high hole barrier at the Pd/WSe₂ interface and the low carrier (electron) concentration in WTe₂. Even if the V_{GS} decreases to the negative direction (bottom-left diagram in Fig. 3c), it is still difficult to inject the electron carriers from WTe₂ to the conduction band of WSe₂ due to the increased electron barrier height caused by the upward bending of the WSe₂ energy band, as well as the low carrier concentration in WTe₂. In contrast, hole carriers are easily injected from Pd to the valence band of WSe₂ because of the reduction in the

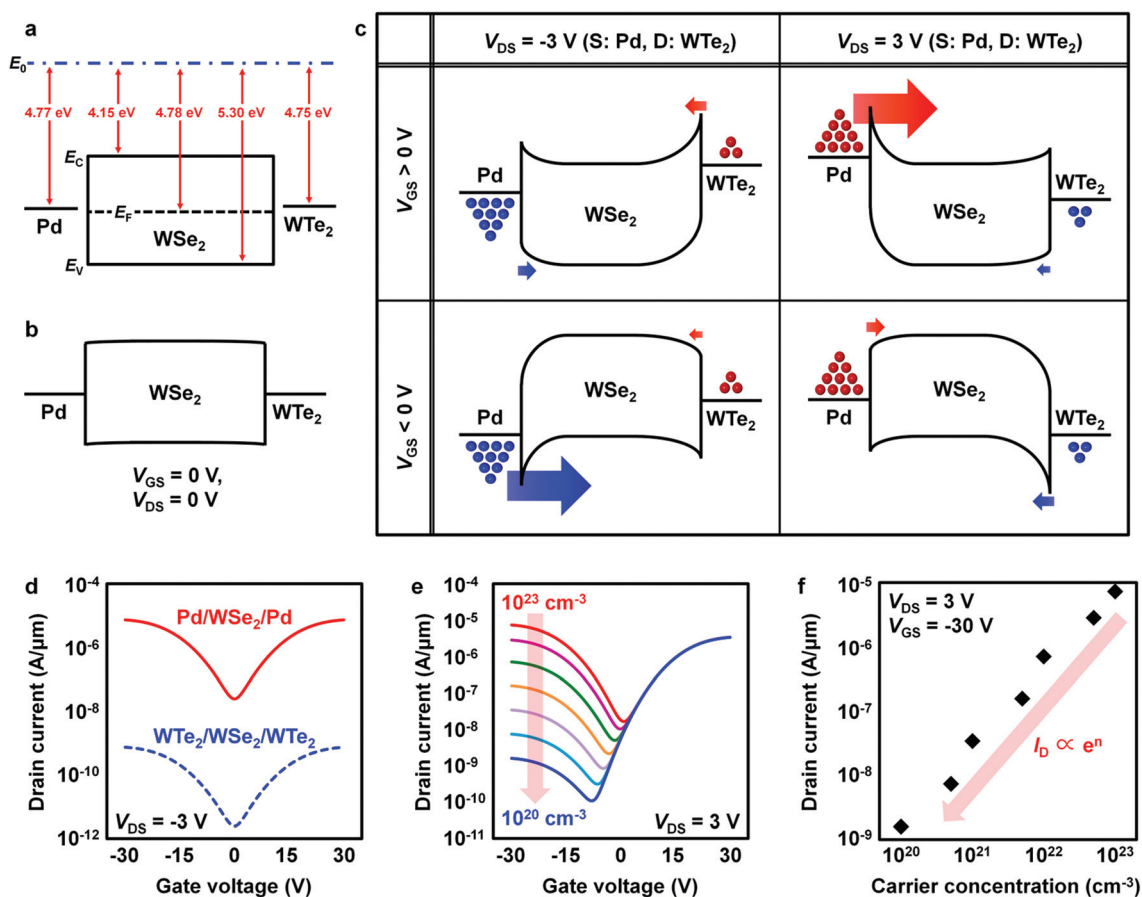


Fig. 3 Predicted energy band diagram in the equilibrium state (a) before and (b) after contact. E_0 , E_C , E_V , and E_F are the vacuum energy level, the lowest energy level of the conduction band, the highest energy level of the valence band, and the Fermi level of the WSe₂, respectively. (c) Energy band diagrams explaining the electron (red) and hole (blue) carrier injection under $V_{DS} = -3$ V and $V_{GS} > 0$ V (top-left diagram), $V_{DS} = -3$ V and $V_{GS} < 0$ V (bottom-left diagram), $V_{DS} = 3$ V and $V_{GS} > 0$ V (top-right diagram), and $V_{DS} = 3$ V and $V_{GS} < 0$ V (bottom-right diagram). The width of the arrows indicates the magnitude of the current. (d) Theoretically calculated I_{DS} – V_{GS} characteristic curves of the Pd/WSe₂/Pd (red solid line) and WTe₂/WSe₂/WTe₂ transistors (blue dashed line), where $V_{DS} = -3$ V. (e) The I_{DS} – V_{GS} characteristic curves of the Pd/WSe₂/WTe₂ transistor, which were calculated under various carrier concentration conditions in WTe₂ (from 10^{23} cm^{−3} to 10^{20} cm^{−3}). (f) The drain current as a function of carrier concentration of WTe₂, which was extracted from the I_{DS} – V_{GS} curves at $V_{DS} = 3$ V and $V_{GS} = -30$ V.

effective hole barrier height of the Pd/WSe₂ junction. Therefore, in the negative- V_{DS} regime, the hole carrier becomes a majority carrier to determine the current of the device, so that the WSe₂/WTe₂ heterojunction transistor exhibits p-type transport behavior, as shown in Fig. 1d. As seen in the right side of Fig. 3c, when positive- V_{DS} and negative- V_{GS} biases are applied (bottom-right diagram), the injections of electron and hole carriers into WSe₂ are hindered by the high electron barrier at the Pd/WSe₂ interface and the low carrier (hole) concentration in WTe₂, respectively. As V_{GS} increases in the positive direction (top-right diagram), the energy band of WSe₂ is bent downward and the effective electron barrier height at the Pd/WSe₂ junction is reduced, subsequently injecting more electron carriers into the conduction band of WSe₂. This probably causes the heterojunction transistor to function as an n-channel device, as mentioned in Fig. 1d. In summary, the polarity of this WSe₂/WTe₂ heterojunction transistor is switched simply by converting the sign of V_{DS} , consequently demonstrating both n- and p-channel transistor operations on single WSe₂ material.

Subsequently, to investigate how the carrier concentration in an electrode affects the heterojunction transistor, we theoretically calculated the thermionic emission current of the transistors formed on Pd/WSe₂/Pd and WTe₂/WSe₂/WTe₂ structures. First of all, we modified the conventional thermionic emission current equation for the heterojunction transistor with WTe₂ electrodes, which is discussed in detail in the ESI.† This modification is required because (i) the current equation is valid under the assumption that the Fermi energy at absolute zero temperature (E_{F0}) is much higher than $k_B T$ ($E_{F0} \gg k_B T$)²⁷ and (ii) high E_{F0} is normally observed in metals with lots of electrons. The predicted I - V characteristic curves of the Pd/WSe₂/Pd and WTe₂/WSe₂/WTe₂ transistors are presented in Fig. 3d. In the case of WTe₂ electrodes, the number of carriers injected from source to channel is less than that of the case using Pd electrodes, consequently decreasing the thermionic emission current from 12 $\mu\text{A } \mu\text{m}^{-1}$ to 2.1 nA μm^{-1} (at $|V_{GS}| = 30$ V). We have also experimentally confirmed that the WTe₂/WSe₂/WTe₂ transistor operates at a much lower current level, which is below 10 nA μm^{-1} over the whole gate voltage range (Fig. S4†). For reference, the Pd/WSe₂/Pd transistor with metal electrodes exhibited ambipolar behavior at a high current level above 1 $\mu\text{A } \mu\text{m}^{-1}$ at $|V_{GS}| = 30$ V (Fig. 1c and Fig. S4b†). We then estimated the electrical characteristic curves of the Pd/WSe₂/WTe₂ heterojunction transistor by varying the carrier concentration of WTe₂ from 10^{23} cm^{-3} to 10^{20} cm^{-3} , as shown in Fig. 3e. As expected, the drain current in the negative gate voltage region shows a decreasing tendency as the carrier concentration in WTe₂ reduces, where a positive- V_{DS} is applied to the device. This is because the hole carrier injection from WTe₂ into the WSe₂ channel region is suppressed by the reduced number of hole carriers in WTe₂. Particularly, because the thermionic emission current depends exponentially on E_{F0} that is proportional to the carrier concentration, the drain current rapidly decays as the carrier concentration of WTe₂ decreases, as seen in Fig. 3f. In addition, we also performed

I_{DS} - V_{GS} measurements in various polarity controllable transistors with different WTe₂ thicknesses to investigate the influence of the WTe₂ thickness on the electrical characteristics of the devices (Fig. S5†). Regardless of the thickness of WTe₂, similar I_{DS} - V_{GS} characteristic curves were observed in the polarity controllable transistors with various WTe₂ thicknesses.

By using this operating principle of the polarity-controllable transistor, we then implemented a complementary inverter, which is a basic building block for logic circuits, on a single WSe₂ channel material. Fig. 4a shows a schematic and an optical image of the inverter circuit fabricated on the WSe₂/WTe₂ heterojunction structure, and its equivalent circuit configuration. The supply voltage (V_{DD}) is applied to one of the electrodes on WSe₂ (top electrode), and the other electrode (bottom electrode) is connected to the ground (V_{SS}). The input voltage (V_{IN}) is supplied through the common back gate terminal (input electrode). The output voltage (V_{OUT}) is then taken at the electrode connected on WTe₂ (output electrode). The V_{IN}

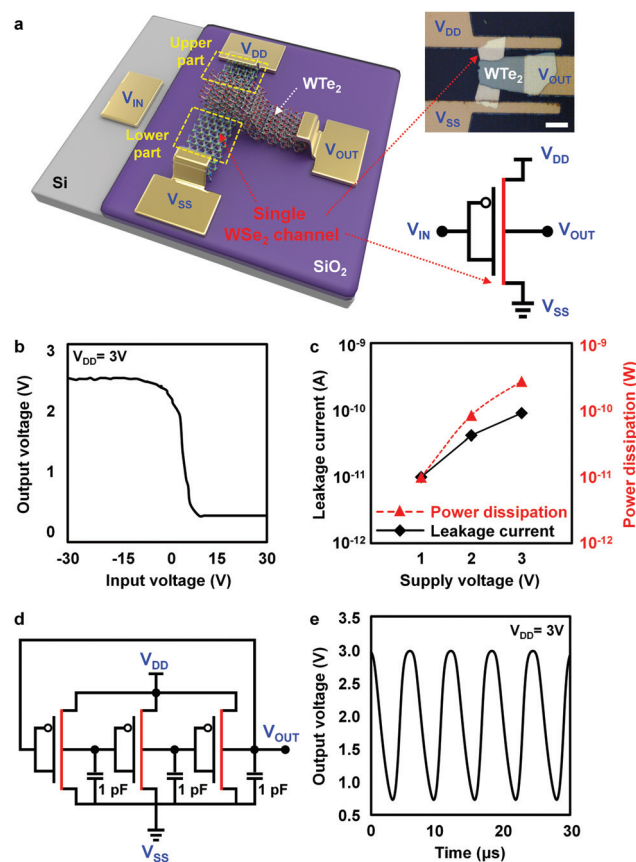


Fig. 4 (a) Three-dimensional schematic (left) and optical image (top-right) of an inverter circuit fabricated on the WSe₂/WTe₂ heterojunction structure and an equivalent inverter circuit configuration (bottom-right). The scale bar represents 10 μm . (b) Voltage-transfer characteristic of the inverter circuit, where $V_{DD} = 3$ V. (c) Leakage current (left axis) and static power consumption (right axis) of the inverter circuit with respect to the supply voltage. (d) Circuit diagram of a three-stage ring oscillator consisting of the inverters. (e) Out signal of the ring oscillator at $V_{DD} = 3$ V.

vs. V_{OUT} characteristic of the inverter is shown in Fig. 4b, where V_{DD} is 3 V. The operation of the inverter can be understood by examining carrier transport in two parts of the WSe_2 channel: (i) the upper part between the top electrode and the output electrode and (ii) the lower part between the output electrode and the bottom electrode, which are indicated by yellow dotted lines in Fig. 4a. This inverter based on the $\text{WSe}_2/\text{WTe}_2$ heterojunction has a single WSe_2 channel, but it has a feature that one of the top and bottom channel parts forms a low resistance path complementarily, according to V_{IN} . When a negative V_{IN} (logic state “0”) is applied, a low-resistance path is formed at the upper part of the channel. This is because the effective hole barrier height at the top Pd electrode/ WSe_2 junction is reduced and the hole carrier easily flows from the top electrode to WTe_2 through the WSe_2 region. Meanwhile, in the lower part, the carrier injection from WTe_2 into the bottom Pd electrode is suppressed due to (i) the low hole concentration of WTe_2 and (ii) the high electron barrier height at the bottom Pd electrode/ WSe_2 junction, thereby inducing a high-resistance path. Therefore, a high voltage (logic state “1”) appears at the output electrode. In contrast, when a positive V_{IN} (logic state “1”) is applied, the upper and lower channel parts form high- and low-resistance paths, respectively. High hole barrier height at the top Pd electrode/ WSe_2 junction and low electron concentration in WTe_2 hinder the injection of carriers from the top and output electrodes into the WSe_2 channel. However, the reduced effective electron barrier height at the bottom Pd electrode/ WSe_2 junction enhances the electron injection from the bottom electrode into the WSe_2 channel. This results in a low resistance path at the lower channel part, yielding a low voltage (logic state “0”) at the output electrode. Particularly, thanks to this complementary operation, the power consumption of this inverter is expected to be significantly low. We estimated a low leakage current of $89 \text{ pA } \mu\text{m}^{-1}$ and a static power consumption of 267 pW for V_{DD} of 3 V (Fig. 4c).

In addition, we demonstrated a three-stage ring oscillator consisting of the inverters based on the polarity-controllable transistor (Fig. 4d). The ring oscillator was modeled on basis of experimentally obtained inverter characteristics in PSpice circuit simulator. Each inverter stage is connected with a load capacitor of 1 pF to avoid unwanted signal noise and rapid voltage spikes. Fig. 4e shows the output signal of the ring oscillator, which was obtained at $V_{\text{DD}} = 3 \text{ V}$. The maximum output voltage swing and the oscillation frequency were 2.2 V and 167 kHz, respectively. In complementary metal oxide semiconductor (CMOS) technology, a ring oscillator requires an odd number of inverters consisting of two p- and n-channel transistors (*i.e.*, spatially separated n- and p-channel layers are required). However, this ring oscillator utilizing the operating principle of the polarity-controllable transistor is expected to be integrated by a simpler fabrication process, compared to the same function circuit *via* conventional CMOS technology. It is because the WTe_2 allows selective carrier injection at source/channel junction so that n- and p-channel layers do not need to spatially separate in the complementary inverter circuit configuration.

Conclusions

We reported a polarity-controllable TMD transistor that operates as both an n- and a p-channel transistor by simply converting the sign of V_{DS} . Unlike typical ambipolar transistors, which cannot be completely turned off due to their continuous transition from the electron-conduction to the hole-conduction state, our polarity-controllable transistor showed low leakage currents of a few picoamperes ($<10 \text{ pA } \mu\text{m}^{-1}$). To understand the carrier-transport mechanism in this polarity-controllable transistor, we performed various analyses, such as Hall effect measurement, KPFM measurement, and temperature-dependent electrical measurement. The carrier concentration of WTe_2 obtained from the Hall measurement was $1.05 \times 10^{20} \text{ cm}^{-3}$, which was significantly higher than the carrier concentrations reported in typical TMD materials (10^{15} – 10^{19} cm^{-3}), and lower than the free-electron concentrations in metals (10^{22} – 10^{23} cm^{-3}). Through the KPFM and temperature-dependent electrical measurements, we predicted the energy band diagram for the Pd/ $\text{WSe}_2/\text{WTe}_2$ heterojunction and understood the operating principle of the polarity-controllable transistor formed on the heterojunction. In this device, by using the metal-like WTe_2 with a low carrier concentration as a drain electrode, we could enable selective carrier injection at the Pd/ WSe_2 junction. This technique is expected to be applicable to other TMD transistors exhibiting ambipolar behavior. Furthermore, based on the operating principle of the polarity-controllable transistor, we demonstrated a homogeneous complementary inverter on a single WSe_2 channel material. Its power consumption was predicted to be the very low value of 267 pW for V_{DD} of 3 V, where the estimated leakage current was $89 \text{ pA } \mu\text{m}^{-1}$. Finally, we confirmed the expandability of this polarity-controllable transistor toward complex circuits by presenting the proper operation of a three-stage ring oscillator consisting of the inverters based on the transistor. The demonstration of the ring oscillator consisting of the polarity-controllable transistor indicates that the complexity of circuit design and fabrication could be simplified.

Author contributions

All authors have given approval to the final version of the manuscript.

Conflicts of interest

There are no conflicts to declare.

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